

Objective

Create a basic cross-platform command execution application that can be installed on Windows and macOS, capable of executing simple system commands and returning their results.

Core Requirements (2 hours)

1. Installation Package

- Create a simple installer for either Windows (.exe) or macOS (.pkg) [choose one]
- Implement basic start-on-boot functionality

2. Command Execution

Implement support for two command types:

- Network ping
- Get system info (hostname, IP address)

```
type Commander interface {  
    Ping(host string) (PingResult, error)  
    GetSystemInfo() (SystemInfo, error)  
}  
  
type PingResult struct {  
    Successful bool  
    Time       time.Duration  
}  
  
type SystemInfo struct {  
    Hostname string  
    IPAddress string  
}
```

3. Communication

- Implement a basic HTTP server to receive commands
- Return results as JSON

Example endpoint structure:

```
// POST /execute  
type CommandRequest struct {
```

```

        Type    string `json:"type"`    // "ping" or "sysinfo"
        Payload string `json:"payload"` // For ping, this is the host
    }

    type CommandResponse struct {
        Success bool    `json:"success"`
        Data    interface{} `json:"data"`
        Error   string    `json:"error,omitempty"`
    }

```

Implementation Requirements

1. Main Application Structure

```

func main() {
    commander := NewCommander()
    server := &http.Server{
        Addr:    ":8080",
        Handler: handleRequests(commander),
    }
    log.Fatal(server.ListenAndServe())
}

func handleRequests(cmdr Commander) http.Handler {
    mux := http.NewServeMux()
    mux.HandleFunc("/execute", handleCommand(cmdr))
    return mux
}

func handleCommand(cmdr Commander) http.HandlerFunc {
    return func(w http.ResponseWriter, r *http.Request) {
        // Parse request and execute command
    }
}

```

2. Platform-Specific Implementation

```

type commander struct{}

func NewCommander() Commander {
    return &commander{}
}

func (c *commander) GetSystemInfo() (SystemInfo, error) {
    hostname, err := os.Hostname()
    if err != nil {
        return SystemInfo{}, err
    }

    // Get IP address (implement this)

    return SystemInfo{
        Hostname: hostname,
        IPAddress: "implement me",
    }
}

```

```
    }, nil  
}
```

Deliverables

1. Create a GitHub repository with your solution.
2. Source code with:
 - Command execution implementation
 - HTTP server implementation
 - Basic installer script
3. README.md with:
 - Build instructions
 - API documentation
 - Installation guide
 - Testing
 - A short clip that shows the "app in action"

Example Test Case

```
func TestGetSystemInfo(t *testing.T) {  
    cmdr := NewCommander()  
    info, err := cmdr.GetSystemInfo()  
  
    if err != nil {  
        t.Fatalf("Expected no error, got %v", err)  
    }  
  
    if info.Hostname == "" {  
        t.Error("Expected hostname to be non-empty")  
    }  
  
    if info.IPAddress == "" {  
        t.Error("Expected IP address to be non-empty")  
    }  
}
```